```c
#include <stdio.h>

int main()
{
    int n, m, i, j, k;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    printf("Enter the number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], request[n][m];
    int avail[m];

    printf("Enter the allocation matrix:\n");

    for(i = 0; i < n; i++)
    {
        printf("Process %d: ", i);

        for(j = 0; j < m; j++)
        {
            scanf("%d", &alloc[i][j]);
        }
    }

    printf("Enter the request matrix:\n");

    for(i = 0; i < n; i++)
    {
        printf("Process %d: ", i);

        for(j = 0; j < m; j++)
        {
            scanf("%d", &request[i][j]);
        }
    }

    printf("Enter the available resources: ");

    for(i = 0; i < m; i++)
    {
        scanf("%d", &avail[i]);
    }

    int finish[n], safeSeq[n], work[m];

    for(i = 0; i < m; i++)
    {
        work[i] = avail[i];
    }
```

```c
for(i = 0; i < n; i++)
{
    finish[i] = 0;
}

int count = 0;

while(count < n)
{
    int found = 0;

    for(i = 0; i < n; i++)
    {
        if(finish[i] == 0)
        {
            int possible = 1;

            for(j = 0; j < m; j++)
            {
                if(request[i][j] > work[j])
                {
                    possible = 0;
                    break;
                }
            }

            if(possible)
            {
                for(k = 0; k < m; k++)
                {
                    work[k] += alloc[i][k];
                }

                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }

    if(found == 0)
    {
        break;
    }
}

if(count == n)
{
    printf("\nSystem is in safe state.\n");
    printf("Safe Sequence is: ");

    for(i = 0; i < n; i++)
    {
        printf("P%d ", safeSeq[i]);
    }
}
else
{
    printf("\nDeadlock detected.\n");
}

return 0;
```

```
Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0
System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1
```